

COMPUTER VISION ASSESSMENT

1. You load an image with `cv2.imread()` and print its shape, getting (480, 640, 3). What do these three values represent?

Ans. The shape (480, 640, 3) represents :

- ⇒ 480 : height of the image in pixels
- ⇒ 640 : width of the image in pixels
- ⇒ 3 : number of color channels (Blue, Green, Red)

2. `cv2.imread()` loads images in BGR order by default. Why does the BGR vs RGB distinction matter when displaying or passing images to other libraries?

Ans. The BGR vs RGB distinction matters because different libraries interpret colour channels differently.

⇒ OpenCV uses BGR format, while libraries like Matplotlib and PIL use RGB format. If the image is passed without conversion, colours appear incorrect.

- For example: Red may appear blue
- Blue may appear red

Therefore, conversion is required before displaying or processing images in other libraries.

3. You resize the same image using INTER_LINEAR and INTER_AREA interpolation. When is INTER_AREA the preferred choice over INTER_LINEAR?

Ans. INTER_AREA is preferred when reducing or shrinking the size of an image because it minimizes distortion and aliasing effects.

- » It works by calculating pixel area relations, producing smoother and better-quality results during downscaling.
- » INTER_LINEAR is generally preferred for enlarging images.

4. You convert an image from BGR to HSV before processing. What does the HSV colour space represent that BGR does not?

Ans. HSV represents colour information in a way closer to human perception.

HSV stands for:

- Hue » actual colour
- Saturation » intensity/purity of colour
- Value » brightness

Unlike BGR, HSV separates colour information from brightness, making it easier for tasks like colour detection and segmentation.

5. You need to extract a specific region from a medical scan. What is the conceptual difference between cropping an image and resizing it?

Ans. Cropping removes unwanted portions of an image and keeps only a selected region.

- » Resizing changes the dimensions of the entire image without removing any part of it.
- » Cropping focuses on region selection, while resizing focuses on changing image size.